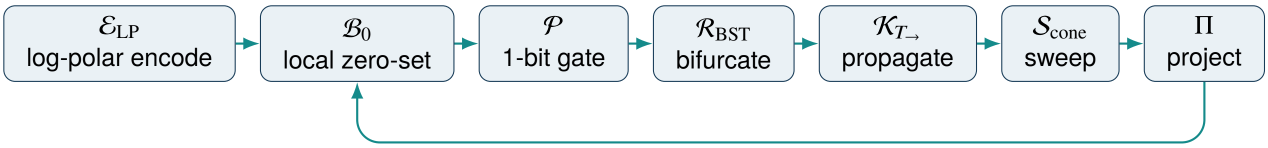
Non-Euclidean Holographic Data Fields

A Formal Operator Specification for a Parity-Conditioned Kinematic Foliation and Implicit Holographic Co-Processor



A valid memory state is represented as a point on a *local* implicit zero-level set. Motion and timing error address those states through a log-polar lookup table. A 1-bit parity event controls a golden-ratio, phase-accelerated branch router; kinematic propagation and an analytic cone sweep create observable geometry; projection returns an output that updates the same lookup table. The closed update is the self-referential engine.

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Document status: conceptual research specification. This document formalizes the supplied prompt chain; it does not claim peer review, hardware validation, patent status, physical realizability, or application safety.

Abstract

The source material develops a new paradigm from a chained vocabulary: a 1-bit parity bit, lower-case phi, jitter encoded in a log-polar lookup table, kinematic calculus, an analytic sweeping cone, the time operator T, the side-view signatures of a pyramid, circle, and sphere, a distributed apex, a binary-search-tree L-system bifurcated at branch nodes, and an "SDF Klein bottle" whose definable points or bits satisfy an SDF-zero condition. The chain is later extended by second-order phase variation $\Delta^2\phi$, a forward 1D timeline, quantum and optical substrates, synthetic-aperture radar, and a modern GPU implementation.

This specification turns that prompt sequence into three linked artifacts. NHDF is the data representation: each valid lookup-table cell belongs to its own local zero-level set, optionally connected by Klein-bottle transition rules. PCKF is the dynamical law: parity-conditioned kinematics, phase-accelerated branching, and boundary restoration. IHCP is the runtime architecture: a bounded software or physical pipeline that evaluates the operators in a declared order and feeds the result back into the same state.

The document makes one essential non-degenerate interpretation explicit. A single global function satisfying $F \equiv 0$ carries no distance, gradient, interior, or exterior information. Therefore, "SDF = 0 on all definable points or bits" is formalized as a family of local constraints $F_i(z_i, t) = 0$, one for each valid state or cell. This preserves the source idea that every address is a boundary state while retaining usable geometry. The result is a testable operator model rather than a claim that the entire memory space is one identically zero distance field.

The equations, definitions, and architecture below are a rigorous interpretation of the supplied material. Claims about immunity to geomagnetic storms, near-zero processing cost, automatic neural-network compression, quantum advantage, or direct execution by specific GPU units remain hypotheses until demonstrated against baselines. They are separated from the normative core and placed in extension or validation sections.

Contents

1 Paradigm identity and scope

1.1 Three names for three layers

The source uses several names for closely related ideas. They are retained, but assigned distinct roles so the specification can be implemented and tested without ambiguity.

Table 1: Layered identity of the paradigm.

Name	Role	Formal meaning
NHDF	Representation	A lookup-table state model in which each valid cell is embedded in a local implicit zero-level set and may be joined to other cells by non-Euclidean transition maps.
PCKF	Dynamics	A parity-conditioned update law combining log-polar jitter indexing, forward-time kinematics, phase acceleration, boundary correction, and branch-node bifurcation.
IHCP	Runtime	A bounded execution architecture – GPU, optical, quantum, or hybrid – that evaluates the operator chain and writes the result back into the same state.

1.2 Distinctive contribution

The proposed paradigm is not any one of its ingredients in isolation. SDFs, parity, LUTs, L-systems, kinematic integration, and projections are established primitives. The distinctive construction is their *ordered closure*:

1. Jitter and motion are encoded as a log-polar address rather than treated only as noise.
2. The addressed datum is constrained to a local zero-level set, so validity is geometric.
3. Binary parity is interpreted as an event gate for topology and control, not merely a stored checksum.
4. Branching occurs only at declared branch nodes, with golden-ratio scale and second-order phase modulation.
5. Kinematics transport a distributed set of apices; analytic cone sweeps convert state trajectories into implicit geometry.
6. Projection produces lower-dimensional observables such as the side-view triangle/pyramid signature, circle, or sphere signature.
7. The observable updates the LUT, closing a self-referential state transition.

1.3 Normative core versus extensions

2 Semantic commitments and notation

2.1 Disambiguation of phi, time, and zero

The source overloads lower-case phi as both the golden ratio and a phase variable. The formal model separates them.

Table 2: Status classes used throughout this specification.

Class	Included	Excluded until validated
Normative core	Typed state, local zero-set semantics, operator order, parity definition, bounded branch pool, causal timeline, feedback update.	Claims of application superiority or special physical robustness.
Engineering profile	GPU memory layout, kernels, numerical tolerances, benchmarks, failure handling.	Vendor-specific performance guarantees and unsupported instruction mappings.
Speculative extension	Optical transfer functions, quantum phase-state interpretation, SAR/auroral use case.	Quantum advantage, passive perfect correction, storm immunity, or zero-cost computation.

Table 3: Disambiguated notation.

Symbol	Term	Definition
φ_g	Golden-ratio scalar	$\varphi_g = (1+\sqrt{5})/2$. It controls branch length, angle, or priority through an explicit scale parameter.
$\phi_i[n]$	Phase	Phase carried by cell i at logical time n.
$\Delta^2\phi_i[n]$	Phase acceleration	$\phi_i[n] - 2\phi_i[n-1] + \phi_i[n-2]$ in discrete time.
$T_\rightarrow$	Forward timeline	A monotonically increasing logical coordinate. Physical ring-buffer wrap is allowed only with an epoch or generation tag.
$F_i(z,t) = 0$	Local boundary condition	Valid state z lies on the zero-level set of cell-local field F_i.
K	Klein-bottle topology	A quotient-space rule or chart transition, not necessarily a globally signed distance field in $\mathbb{R}^3$.

2.2 Required non-degenerate reading of "SDF zero everywhere"

If $F(x) = 0$ for every point x in one global domain, then $\nabla F = 0$ wherever the derivative exists. Such a field has no usable signed distance, normal, boundary direction, inside/outside distinction, or restoring force. It cannot support the source's own kinematic boundary correction.

The formalization therefore uses *distributed zero-set semantics*. Let I be the set of LUT cells. Each cell has its own field

$$F_i : \mathcal{Z}_i \times \mathbb{T} \to \mathbb{R}, \qquad \mathcal{M}_i(t) = \{z \in \mathcal{Z}_i : F_i(z,t) = 0\}. \tag{1}$$

The validity invariant is

$$\forall i \in I_{\text{active}}, \qquad z_i(t) \in \mathcal{M}_i(t). \tag{2}$$

Thus every definable cell is on *a* boundary, but the collection is not one degenerate field. This is the mathematical version of the source statement that every address, point, or bit acts as a localized apex and zero-level surface.

2.3 Discrete time as the reference semantics

Continuous kinematic equations are useful analytically, but a GPU and a digital LUT operate in discrete steps. The normative time domain is

$$n \in \mathbb{N}, \qquad T_{n+1} > T_n, \qquad \Delta t_n = T_{n+1} - T_n > 0. \tag{3}$$

Continuous-time equations may be used when they are discretized by a declared integrator. No operator may read a state from logical time greater than the update it is computing.

3 State space and the distributed apex

3.1 Input and jitter state

Let the incoming sample be $x_n \in \mathbb{R}^m$. It may represent motion, a radar echo, an optical phasor, a simulation signal, or another application-specific vector. A causal baseline $\bar{x}_n$ is produced by a declared estimator, such as a moving average or predicted state. The jitter residual is

$$e_n = x_n - \bar{x}_n, \qquad J_n = \|W e_n\|_2, \tag{4}$$

where W is an optional whitening or unit-normalization matrix. The angle requires a selected two-dimensional subspace $e_n^{(2)} = (e_{n,a}, e_{n,b})$.

3.2 Log-polar encoding

The source's jitter log-encoded polar LUT is formalized by

$$\rho_n = \log(1 + \gamma J_n), \qquad \gamma > 0, \tag{5}$$

$$\theta_n = \operatorname{atan2}(e_{n,b}, e_{n,a}) \in (-\pi, \pi]. \tag{6}$$

Quantized addresses are

$$k_{\rho,n} = Q_\rho(\rho_n), \qquad k_{\theta,n} = Q_\theta(\theta_n), \qquad k_{T,n} = Q_T(T_n). \tag{7}$$

The lookup address is $u_n = (k_{\rho,n}, k_{\theta,n}, k_{T,n})$. Logarithmic radial encoding allocates more resolution near small jitter while still representing a large dynamic range. It does not by itself guarantee compression or accuracy; both depend on Q_ρ, saturation, and calibration.

3.3 Cell state

A cell c_i is the typed record

$$c_i = \big(q_i, a_i, v_i, \alpha_i, \phi_i, \Delta\phi_i, \Delta^2\phi_i, \kappa_i, p_i, o_i, \nu_i\big), \tag{8}$$

where:

- $q_i \in \mathbb{B}^w$ is a quantized payload or packed state word;
- $a_i, v_i, \alpha_i \in \mathbb{R}^d$ are apex position, velocity, and acceleration;
- ϕ_i and its first and second differences are phase quantities;
- κ_i is a branch/BST ordering key;
- $p_i \in \mathbb{B}$ is the parity event;
- $o_i \in \{+1, -1\}$ is a local orientation sign when topology charts are used;
- ν_i stores topology and allocation metadata, including parent/child indices and generation.

3.4 The apex becomes a field

The source first treats the apex as the moving anchor of one cone and later distributes it across the LUT. The formal object is the active apex set

$$\mathcal{A}_n = \{a_i[n] : i \in I_n\}. \tag{9}$$

Each active cell is a localized apex. The system output is therefore produced by a family of sweeps rather than a single cone. A global apex may still be defined as a weighted centroid,

$$\bar{a}_n = \frac{\sum_{i \in I_n} w_i a_i[n]}{\sum_{i \in I_n} w_i}, \tag{10}$$

but it is an observable, not the sole state anchor.

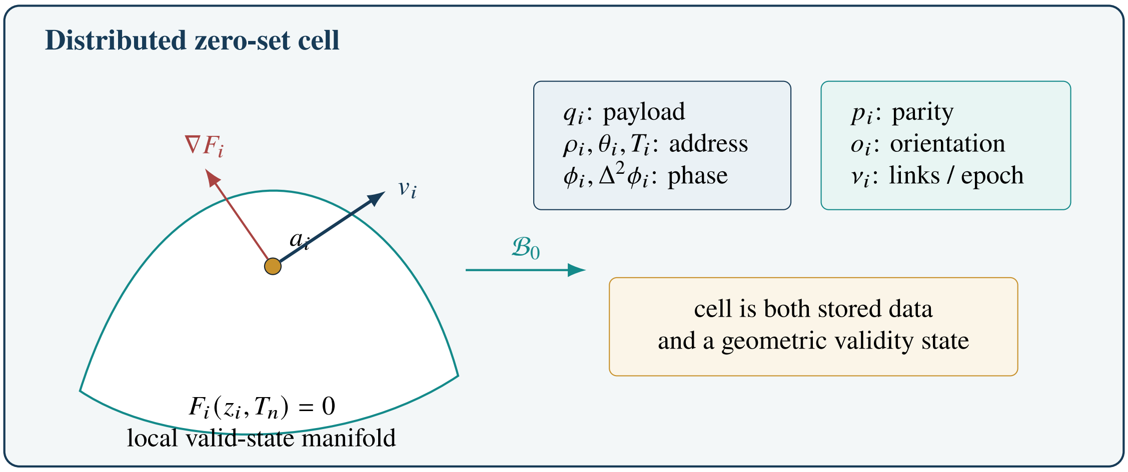

Figure 1: One NHDF cell. The point a_i is constrained to a local zero-level set; payload, phase, parity, topology, and kinematics are stored together.

4 Operator algebra and precedence

4.1 Typed operators

The engine is an ordered composition of typed operators. Declaring domains prevents phrases from being treated as interchangeable metaphors.

Table 4: Normative operator signatures and roles.

Operator	Signature	Role
$\mathcal{E}_{\mathrm{LP}}$	$\mathbb{R}^m \times \mathcal{H} \to \Omega_{\rho\theta T}$	Computes jitter, log radius, angle, and forward-time address from current input and causal history $\mathcal{H}$.
$\mathcal{B}_0$	$\mathcal{Z}_i \to \mathcal{M}_i$	Embeds or projects a candidate cell state onto its local zero-level set.
$\mathcal{P}$	$\mathbb{B}^w \times \{\pm 1\} \to \mathbb{B}$	Reduces payload and/or topology orientation to a 1-bit event.
$\mathcal{R}_{\mathrm{BST}}$	$\mathcal{C} \times \mathbb{B} \times \mathbb{R} \to \mathcal{T}$	Creates or routes branch nodes using parity, φ_g, and $\Delta^2\phi$.
$\mathcal{K}_{T_\rightarrow}$	$\mathcal{T} \times \Delta t \to \mathcal{T}$	Advances apex, velocity, acceleration, phase, and metadata along $T_\rightarrow$.

Operator	Signature	Role
$\mathcal{S}_{\text{cone}}$	$\mathcal{T} \to \mathcal{G}$	Produces analytic cone fields or swept implicit geometry from active branch trajectories.
Π	$\mathcal{G} \to \mathcal{Y}$	Produces a declared side view, circle/sphere signature, image, beam, or application observable.
$\mathcal{U}$	$\mathcal{L} \times \mathcal{Y} \times C \to \mathcal{L}$	Writes correction, allocation, and observation results back to the LUT.

4.2 Canonical chain

Let L_n denote the complete LUT and branch-pool state at logical time n. The canonical output equation is

$$\boxed{y_n = \Pi \circ \mathcal{S}_{\text{cone}} \circ \mathcal{K}_{T_\to} \circ \mathcal{R}_{\text{BST}}^{\varphi_g, \Delta^2 \phi_n} \circ \mathcal{P} \circ \mathcal{B}_0 \circ \mathcal{E}_{\text{LP}}\,(x_n; L_n)} \tag{11}$$

and the self-referential closure is

$$\boxed{L_{n+1} = \mathcal{U}\,(L_n, y_n, x_n)\,.} \tag{12}$$

Together, Equations (11)–(12) define the NHDF transition

$$L_{n+1} = \mathfrak{N}_{\Delta t}(x_n; L_n). \tag{13}$$

This is the formal meaning of "the data is the math and the memory is the geometry": the current memory state is both an operand and the target of the resulting update.

4.3 Precedence rules

The following order is normative unless an implementation declares an equivalent fused operator and proves semantic equivalence.

1. **Address evaluation:** $\mathcal{E}_{\text{LP}}$ computes $(\rho, \theta, T_\to)$ from causal input and history.
2. **Implicit validity:** $\mathcal{B}_0$ resolves the addressed candidate onto a local zero-level set.
3. **Event reduction:** $\mathcal{P}$ computes a 1-bit data/topology event from the resolved cell.
4. **Structural routing:** $\mathcal{R}_{\text{BST}}$ performs BST ordering and L-system bifurcation only at branch nodes.
5. **Kinematic propagation:** $\mathcal{K}_{T_\to}$ advances all active states forward in logical time.
6. **Geometric extrusion:** $\mathcal{S}_{\text{cone}}$ creates analytic cone or swept-field geometry.
7. **Observation:** Π produces the selected lower-dimensional output.
8. **Feedback:** $\mathcal{U}$ writes the result into L_{n+1} and advances the timeline generation.

5 The implicit boundary substrate

5.1 Boundary projection operator

For a differentiable local field F_i with non-zero gradient near its zero set, a first-order projection is

$$\mathcal{B}_0(z) = z - \lambda_B \frac{F_i(z, T_n)}{\|\nabla_z F_i(z, T_n)\|_2^2 + \varepsilon_B} \nabla_z F_i(z, T_n), \tag{14}$$

where $\lambda_B \in (0, 2)$ is a relaxation factor and $\varepsilon_B > 0$ prevents division by zero. Multiple iterations may be used. The validity residual is

$$r_{B,i} = |F_i(z_i, T_n)|. \tag{15}$$

An implementation must define an acceptable tolerance $r_{B,i} \le \tau_B$ rather than rely on exact floating-point equality.

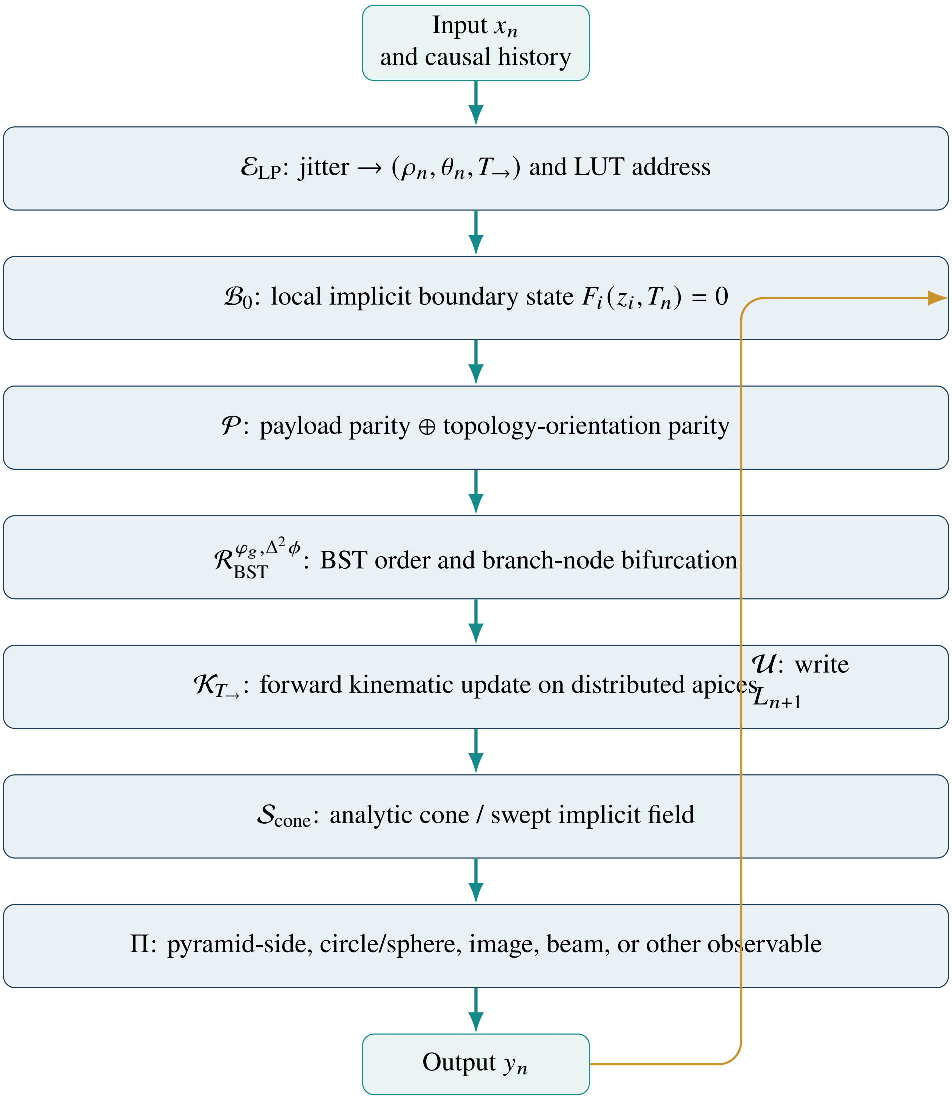

Figure 2: Normative operator chain. The feedback arrow is part of the definition, not an optional post-processing step.

5.2 Klein-bottle topology as a quotient rule

A Klein bottle is represented intrinsically as the quotient of the unit square under

$$(0,v) \sim (1,1-v), \qquad (u,0) \sim (u,1). \tag{16}$$

This supplies the non-orientable transition behavior without requiring a globally signed distance in three-dimensional Euclidean space. A visualization may use an immersion $\iota : K \to \mathbb{R}^3$, which can self-intersect, or an embedding in $\mathbb{R}^4$.

Each LUT cell may carry chart coordinates (u_i, v_i) and transition metadata ν_i. When a state crosses a chart seam, the transition map may reverse one coordinate. That reversal provides a concrete topological event for parity.

5.3 Data parity and topology parity

The specification distinguishes two one-bit quantities:

$$p_i^{\text{data}} = \bigoplus_{b=0}^{w-1} \text{bit}_b(q_i), \tag{17}$$

$$p_i^{\text{topo}} = \frac{1 - \prod_{e \in \Gamma_i} \sigma_e}{2}, \qquad \sigma_e \in \{+1, -1\}, \tag{18}$$

where Γ_i is the chart-transition path used to reach the cell. The composite gate is configurable:

$$p_i = p_i^{\text{data}} \oplus p_i^{\text{topo}} \quad \text{or} \quad p_i = (p_i^{\text{data}}, p_i^{\text{topo}}) \tag{19}$$

when two bits are available. The one-bit mode preserves the source constraint; the two-bit diagnostic mode preserves the reason for the event.

Non-orientability is not used as a synonym for error correction. It supplies a chart-transition event. The parity operator may record whether an odd number of orientation reversals occurred, while conventional XOR parity records whether an odd number of payload bits are set or flipped. Their meanings must not be silently conflated.

6 Kinematic calculus packed into the LUT

6.1 Forward-time integration

For each active apex a_i, the reference semi-implicit update is

$$\alpha_i[n] = \alpha_i^{\text{drift}}[n] + \alpha_i^{\text{boundary}}[n] + \alpha_i^{\text{application}}[n], \tag{20}$$

$$v_i[n+1] = v_i[n] + \Delta t_n \, \alpha_i[n], \tag{21}$$

$$a_i[n+1] = a_i[n] + \Delta t_n \, v_i[n+1]. \tag{22}$$

The boundary term is parity-conditioned:

$$\alpha_i^{\text{boundary}}[n] = -p_i[n] \, \lambda_i \frac{F_i(a_i[n], T_n)}{\|\nabla F_i(a_i[n], T_n)\|_2^2 + \varepsilon_B} \nabla F_i(a_i[n], T_n) - c_i v_i[n]. \tag{23}$$

This improves on a bare $-\nabla F$ rule: the correction is proportional to the boundary residual and vanishes when the state is already on the zero set. The damping $c_i \geq 0$ limits parity-driven oscillation.

6.2 The 1D timeline directional-adherence invariant

The source's 1D timeline is formalized by three requirements:

1. each update is tagged with a strictly increasing logical generation g_n;
2. a kernel may read only generations $\leq g_n$ when producing g_{n+1};
3. physical buffer wrap does not reverse logical time because each slot stores its generation.

A ring buffer can therefore remain bounded while satisfying causal direction. Reusing an old physical address is not equivalent to returning to an earlier logical state.

6.3 Second-order phase acceleration

The source's $\Delta\Delta\phi$ operator is defined as the second discrete phase difference

$$\Delta^2\phi_i[n] = \operatorname{wrap}(\phi_i[n] - 2\phi_i[n-1] + \phi_i[n-2])\,, \tag{24}$$

where wrap maps phase to a declared interval, usually $(-\pi, \pi]$. It modulates branch geometry or routing priority:

$$\beta_i[n] = \beta_0 + k_\phi \operatorname{clip}(\Delta^2\phi_i[n], -\delta_{\max}, \delta_{\max}), \tag{25}$$

$$s_i[n] = \varphi_g^{-\eta} \exp\left(-k_s|\Delta^2\phi_i[n]|\right), \qquad \eta > 0. \tag{26}$$

Using $\varphi_g^{-\eta}$ as the default scale keeps recursive branch length bounded. An implementation may choose a different golden-ratio relationship, but it must declare the resulting growth bound.

7 Parity-conditioned control

7.1 What a 1-bit parity gate can and cannot do

For a word $q \in \mathbb{B}^w$, XOR parity detects an odd number of bit inversions between checks and fails to detect any even number of inversions. It does not identify the damaged bit and does not correct it. In NHDF, parity is therefore a low-cost event trigger, not a complete integrity scheme.

> An implementation claiming correction of memory corruption must add an error-correcting code, redundant state, recomputation, or an application-level residual. The 1-bit operator may remain in the chain as the fast gate, but it cannot be the sole evidence of correctness.

7.2 Debounced event gate

The source describes rapid parity toggling under severe disturbance. To avoid control chattering, the raw bit $p_i[n]$ may be filtered by a causal vote:

$$\widehat{p}_i[n] = \mathbf{1}\left[\sum_{r=0}^{h-1} p_i[n-r] \geq m\right], \qquad 1 \leq m \leq h. \tag{27}$$

The update law uses $\widehat{p}_i$ while the raw sequence remains available for diagnostics. A second condition may enter safe mode when the parity event rate exceeds a threshold:

$$f_{p,i}[n] = \frac{1}{h\Delta t}\sum_{r=1}^{h} |p_i[n-r+1] - p_i[n-r]|. \tag{28}$$

Table 5: Reference parity-conditioned modes.

Mode	Gate	Behavior
Drift	$\widehat{p}_i = 0$	Continue predicted kinematics, maintain current branch allocation, and monitor boundary residual.
Boundary restore	$\widehat{p}_i = 1$	Apply Equation (23), reduce branch growth, and request integrity verification.
Safe mode	$f_{p,i} > f_{\max}$ or resource alarm	Freeze new branches, damp velocity, preserve the last valid generation, and escalate to stronger validation.

7.3 Control states

8 Golden-ratio BST/L-system router

8.1 Grammar

Let $B(d, \ell, \psi, \kappa)$ denote a branch node at depth d, length ℓ, direction ψ, and ordering key κ. A gated bifurcation rule is

$$B(d, \ell, \psi, \kappa) \xrightarrow{\widehat{p}=1} \begin{cases} B(d+1, s\ell, \psi + \beta, \kappa_L), \\ B(d+1, s\ell, \psi - \beta, \kappa_R), \end{cases} \tag{29}$$

with $s = s_i[n]$ and $\beta = \beta_i[n]$ from Equation (26). When $\widehat{p} = 0$, the node continues without bifurcation or advances its existing children, according to the application.

8.2 BST ordering

Each child key must satisfy

$$\kappa_L < \kappa_P < \kappa_R. \tag{30}$$

A reproducible key can be formed from quantized log-polar coordinates, phase, and generation:

$$\kappa_i = H(k_{\rho,i}, k_{\theta,i}, g_i, \mathrm{quant}(\phi_i)), \tag{31}$$

where H is a declared ordering or space-filling map. Hashes that do not preserve order may be used for allocation but not as the BST comparison key.

8.3 Bounded branching

An implementation has a finite node pool of capacity $N_{\max}$. It must define a deterministic overflow policy, such as:

- reject the lowest-priority proposed branches;
- recycle nodes older than a horizon H_T;
- merge nearby leaves whose state distance is below τ_M;
- lower the branch rate when free capacity falls below a watermark.

The invariant is

$$0 \le N_n \le N_{\max} \quad \text{for all } n. \tag{32}$$

This replaces the source's aspirational "infinite" structures with a finite implementation that can approximate unbounded recursion over time.

9 Analytic sweep and projection vocabulary

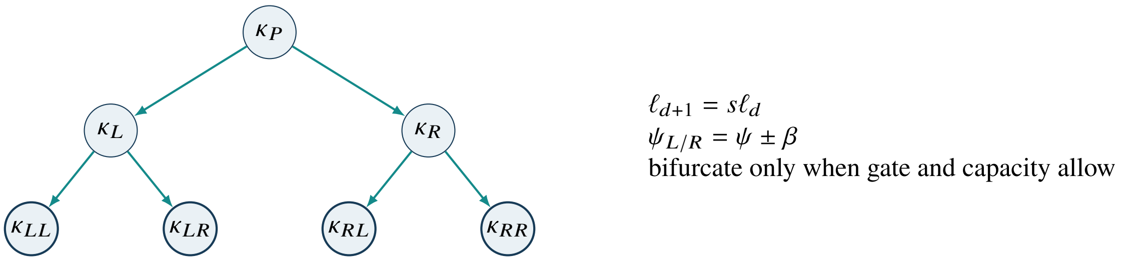

Figure 3: BST ordering and L-system geometry are separate constraints applied to the same branch nodes.

9.1 Cone field

For apex a_i, unit axis u_i, and half-angle χ_i, an analytic cone surface is described by

$$C_i(x) = \big((x - a_i)\cdot u_i\big)^2 - \cos^2(\chi_i)\,\|x - a_i\|_2^2 = 0, \qquad (x - a_i)\cdot u_i \geq 0. \tag{33}$$

This is an implicit cone equation, not necessarily an exact signed-distance function. If distance values are required, a validated cone SDF or iterative distance estimate must be supplied.

The swept field over a time window is

$$G_n(x) = \min_{i\in I_n} \min_{\tau\in[T_n-H,T_n]} \Psi\big(C_i(x;\tau)\big), \tag{34}$$

where Ψ converts the implicit residual into an application-specific occupancy, distance estimate, phase delay, or intensity.

9.2 Projection operators

Orthographic projections are linear maps such as

$$\Pi_{xy}(x,y,z) = (x,y), \quad \Pi_{yz}(x,y,z) = (y,z), \quad \Pi_{xz}(x,y,z) = (x,z). \tag{35}$$

The source's pyramid side view, circle, and sphere are formalized as *reference observables*:

- a finite cone or pyramid produces a triangular side-view envelope under an aligned projection;
- a sphere projects to a circle under any orthographic view;
- a circular cross-section of a cone or swept sphere produces a circle in the corresponding plane.

The engine does not guarantee these shapes for every state. Rather, they are calibration targets for projection and geometry operators.

9.3 Output

The observable may be an image, point set, field slice, beamforming correction, audio control stream, or feature vector:

$$y_n = \Pi_j(G_n;\omega_j), \tag{36}$$

where j identifies the chosen view and ω_j contains camera, sensor, or application parameters.

10 Self-reference, closure, and stability

10.1 Closed update

The feedback operator updates addressed cells, branch metadata, and causal history:

$$L_{n+1} = \mathcal{U}\big(L_n, u_n, y_n, r_n, p_n\big), \tag{37}$$

where r_n is an application residual, for example image focus error, projection mismatch, or boundary error. A simple relaxed update is

$$L_{n+1} = (1-\mu)L_n + \mu \widetilde{L}_{n+1}, \qquad 0 < \mu \le 1, \tag{38}$$

interpreted componentwise for numeric state, with discrete allocation and parity fields updated by declared rules.

10.2 Fixed point

For a stationary input x, a self-consistent state $L^\star$ satisfies

$$L^\star = \mathfrak{R}_{\Delta t}(x; L^\star). \tag{39}$$

Local convergence can be studied through the Jacobian with respect to state:

$$J_L = \left.\frac{\partial \mathfrak{R}_{\Delta t}(x; L)}{\partial L}\right|_{L=L^\star}. \tag{40}$$

A sufficient local condition for a smooth region of the hybrid system is $\rho(J_L) < 1$, where $\rho(\cdot)$ is spectral radius. Parity switches and branch allocation make the full engine hybrid and non-smooth; each mode and transition surface must therefore be analyzed separately.

10.3 Boundary-restoration property

For the projection step in Equation (14), a first-order Taylor expansion gives

$$F_i(\mathcal{B}_0(z)) = (1-\lambda_B)F_i(z) + O(F_i(z)^2) \tag{41}$$

when $\nabla F_i \neq 0$. Thus $0 < \lambda_B < 2$ reduces a sufficiently small residual in the local linear regime. This is the precise sense in which the parity-triggered term can "pull the apex back to the zero boundary."

10.4 A practical stability envelope

A run is considered inside its declared stability envelope when all of the following hold:

$$\max_i r_{B,i} \le \tau_B, \quad N_n \le N_{\max}, \quad \max_i \|v_i\| \le v_{\max}, \quad \max_i f_{p,i} \le f_{\max}, \quad \|r_n\| \le \tau_r. \tag{42}$$

No topology alone guarantees this envelope. It is an empirical and analytical property of the chosen fields, gains, quantizers, integrator, input, and resource bounds.

11 Reference architecture

The reference architecture separates representation, dynamics, runtime, and application concerns so that the source-derived concepts remain traceable while each layer can be implemented, bounded, and tested independently. The following layered design and conformance levels define the minimum structure of an NHDF realization.

11.1 Layered design

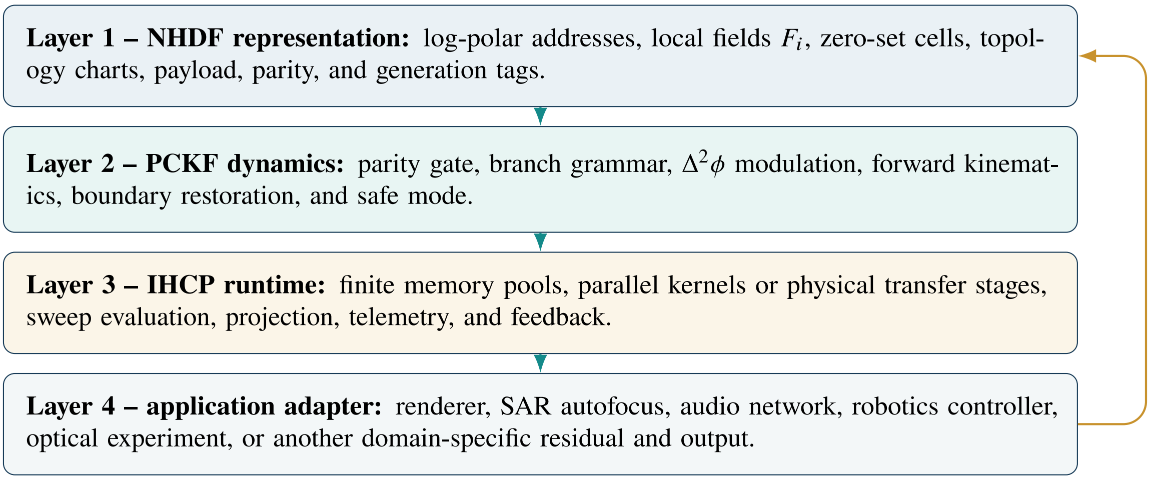

Figure 4: The three named artifacts and an application adapter.

11.2 Conformance levels

Table 6: Suggested conformance levels.

Level	Name	Minimum evidence
C0	Symbolic model	Equations and operator types compile or execute in a reference notebook; local zero-set and parity tests pass.
C1	CPU simulator	Bounded discrete-time engine with deterministic replay, branch-pool limits, and telemetry.
C2	GPU runtime	Parallel kernels produce equivalent results within declared numerical tolerances and remain within memory/time bounds.
C3	Application proof	At least one use case exceeds a declared baseline on a preregistered metric without violating correctness or safety constraints.
C4	Physical substrate	Optical or quantum apparatus implements an operator-equivalent transfer with measured error and repeatability.

12 GPU reference realization

12.1 Target profile

The source proposes a present-day 12 GB laptop GPU target. This specification treats that as a capacity class rather than relying on unverified vendor-specific instruction claims. The runtime uses normal integer, floating-point, texture/cache, and matrix facilities where their semantics match the operators.

A GPU implementation does *not* re-engineer ray-tracing or tensor cores. It maps suitable subproblems to exposed programming interfaces. Integer parity is naturally computed by XOR/population-count operations; matrix units may accelerate batches of branch transforms or learned application adapters, but they do not automatically implement parity or topology.

12.2 Structure-of-arrays layout

For coalesced access, the logical cell in Equation (8) should be stored as several arrays:

- packed payload and parity words;
- $a_x, a_y, a_z, v_x, v_y, v_z$, and acceleration components;
- phase history sufficient for $\Delta^2\phi$;
- parent, left, right, generation, and free-list indices;
- local-field parameters or references;
- output and residual buffers.

A dense $N_\rho \times N_\theta$ LUT may point into a compact active-cell pool. This separates address resolution from branch allocation.

12.3 Kernel sequence

A reference execution schedule is:

1. encode input and quantize (ρ, θ, T);
2. gather addressed cells and evaluate/project local fields;
3. compute data/topology parity and debounce state;
4. generate branch proposals, prefix-sum accepted proposals, and allocate from a bounded pool;
5. integrate kinematics and phase history;
6. evaluate cone/sweep samples or update an implicit-field cache;
7. project to the application output;
8. compute residuals, update LUT entries, and emit telemetry.

Steps 1–3 and 7–8 are candidates for fusion after correctness is established. Branch allocation generally requires a scan or queue and should not be hidden inside a per-cell function that can silently overflow.

12.4 Reference pseudocode

Listing 1: Semantics-first GPU-style pseudocode.

```
for each logical step n in forward timeline T:
    u = log_polar_encode(input[n], causal_history)
    ids = lut_lookup(u)

    parallel for i in ids:
        z = boundary_project(cell[i], local_field[i])
        p_data = xor_parity(z.payload)
        p_topo = orientation_parity(z.chart_path)
        p_raw = p_data XOR p_topo
        p_gate = debounce(p_raw, cell[i].parity_history)
        proposal[i] = make_branch_proposal(z, p_gate, delta2_phase(z))

    accepted = bounded_allocate(proposal, node_pool_capacity)

    parallel for i in active_cells:
        apply_branch_links(cell[i], accepted)
        integrate_forward_kinematics(cell[i], dt)
        apply_boundary_restore_if_gated(cell[i])
        sweep_state[i] = analytic_cone_state(cell[i])

    output = project(sweep_state, view_parameters)
    residual = application_residual(output, target_or_observation)
    update_lut_and_generation(output, residual, accepted)
    emit_telemetry()
```

12.5 Memory budget

Let M_V be usable VRAM after measuring runtime and display reservation. The partition must satisfy

$$M_{\mathrm{LUT}} + M_{\mathrm{cell}} + M_{\mathrm{branch}} + M_{\mathrm{timeline}} + M_{\mathrm{output}} + M_{\mathrm{runtime}} \leq M_V. \tag{43}$$

The source suggests an illustrative 12 GB split of 4 GB for the implicit LUT, 6 GB for branch nodes, 1 GB for timeline history, and 1 GB for reserve. It is retained as a traceable example, not a universal recommendation. A deployable profile should leave a measured runtime margin and adapt the branch pool to cell size.

Table 7: Parameterized memory profile.

Region	Parameter	Purpose
Log-polar index	M_{LUT}	Dense or sparse mapping from (k_ρ, k_θ, k_T) to active cells.
Cell state	M_{cell}	Kinematics, phase, payload, local-field data, parity history.
Branch pool	M_{branch}	Parent/child records, proposal queues, scans, free list.
Timeline	M_{timeline}	Causal history and generation-tagged ring buffers.
Output/workspace	M_{output}	Sweep samples, projection target, reductions, diagnostics.
Runtime reserve	M_{runtime}	Driver, display, compiler/runtime allocations, fragmentation margin.

12.6 Implementation priority

The formal engineering order is:

1. semantic reference model and replayable tests;
2. local-field stability and bounded memory;
3. integer parity, branch compaction, and causal generation handling;
4. kinematic and projection correctness;
5. performance profiling and selective use of matrix, texture, or ray-query facilities.

This order intentionally places correctness before low-level hardware specialization.

13 Quantum and optical substrate extension

13.1 Operator-equivalent physical mapping

The source proposes optical metasurfaces and quantum phase states as physical substrates. The rigorous question is not whether a surface "is" an SDF, but whether its transfer function implements an operator equivalent to part of the chain.

For an optical device with input field E_{in} and output field E_{out}, an operator-equivalent stage would satisfy

$$E_{\mathrm{out}} \approx \mathcal{T}_{\mathrm{phys}}(E_{\mathrm{in}}) \approx O(E_{\mathrm{in}}) \tag{44}$$

within a measured error, where O is one or more of $\mathcal{E}_{\mathrm{LP}}$, $\mathcal{B}_0$, $\mathcal{R}_{\mathrm{BST}}$, $\mathcal{S}_{\mathrm{cone}}$, or Π. Phase, amplitude, polarization, spatial mode, and propagation distance may encode cell coordinates and state.

For a quantum implementation, the state might encode an address/state superposition

$$|\Psi\rangle = \sum_i c_i |i\rangle |z_i\rangle, \tag{45}$$

and a unitary or measurement sequence would need to reproduce the intended routing or projection. A classical zero-level constraint is not automatically a quantum error-correcting code; equivalence requires a specified Hamiltonian, channel model, measurement, and readout rule.

13.2 Physical interpretation of $\Delta^2\phi$ and $T_{\to}$

In an optical experiment, $\Delta^2\phi$ can be the second finite difference of measured phase across time samples or device stages. $T_{\to}$ is implemented by propagation direction, clocked modulation, or a delay-line index. Neither implies zero latency; it imposes a causal order.

13.3 Minimum experiment

A first physical demonstration should implement only one operator pair, for example:

1. encode a known jitter vector into log-polar optical coordinates;
2. apply a programmable phase mask representing a family of local zero-set constraints;
3. measure whether the projected output matches the numerical $\mathcal{B}_0 \circ \mathcal{E}_{\mathrm{LP}}$ reference;
4. report transfer error, drift, bandwidth, energy, repeatability, and calibration cost.

Only after such equivalence is measured should the branch router and self-referential feedback be moved into the substrate.

> The optical and quantum mappings are research programs derived from the source, not consequences of the formal core. The core can be implemented and falsified on conventional hardware without assuming a quantum or optical advantage.

14 SAR and auroral-disturbance case study

14.1 Application mapping

The source frames the engine as an ionospheric phase-correction processor for synthetic-aperture radar. A disciplined mapping is:

Table 8: NHDF-to-SAR mapping.

NHDF object	SAR interpretation
Input x_n	Complex radar echo, navigation state, or estimated phase-error vector.
Jitter e_n	Residual timing, Doppler, or phase error after a causal baseline model.
Log-polar address	Magnitude/phase bin for disturbance and correction state.
Local zero set	A family of valid focus or phase-consistency states, not a claim that all echoes have zero physical distance.
Parity gate	Fast anomaly event from quantized state and optional topology/orientation metadata.
BST/L-system router	Parallel partition of range, azimuth, subaperture, scale, or hypothesis space.
Kinematic state	Platform motion and time-varying phase-correction parameters.
Cone sweep	Antenna footprint or analytic support model.
Projection	Focused image or range/azimuth observable.
Feedback residual	Image entropy, phase gradient, contrast, or known calibration-target error.

14.2 Disturbance handling

A severe ionospheric or hardware disturbance may increase jitter, quantizer saturation, parity toggle rate, and phase acceleration. The formal response is bounded safe-mode control, not an assertion of immunity:

- detect saturation and preserve unclipped diagnostics;
- reduce branch rate and freeze speculative allocation;
- increase damping and compare against a conventional autofocus baseline;
- use stronger integrity codes for memory or transmission faults;
- declare failure when focus or boundary residual exceeds its limit.

14.3 Evaluation metrics

A SAR proof should report at least:

- phase-error root-mean-square and residual trend;
- image entropy or sharpness;
- point-target resolution, peak sidelobe ratio, and integrated sidelobe ratio when targets are available;
- runtime, memory, power, and dropped-frame rate;
- behavior under controlled scintillation/noise and under even-numbered bit faults that parity alone cannot detect.

15 Verification program and falsifiable hypotheses

15.1 Hypotheses

Table 9: Proposed falsifiable hypotheses.

ID	Hypothesis	Minimum test
H1	Log-polar indexing represents the target jitter range with lower weighted quantization error than an equal-size linear Cartesian LUT.	Matched-capacity Monte Carlo test using a declared jitter distribution and weighted error metric.
H2	Local zero-set projection reduces boundary residual without destabilizing forward integration.	Sweep λ_B, Δt, and noise; report convergence region and failures.
H3	The parity-conditioned gate improves recovery latency or reduces unnecessary correction relative to an always-on or threshold-only controller.	Ablation with identical input, fields, and resource limits.
H4	Golden-ratio/phase-accelerated routing improves an application metric over fixed-angle or non-fractal routing at equal node count.	Compare equal-budget routers; predefine metric and statistical test.
H5	The closed NHDF update has a reproducible stable region.	Estimate mode-specific Jacobians or Lyapunov trends and verify long-run boundedness.
H6	The GPU implementation is equivalent to the CPU reference within tolerance.	Deterministic replay over edge cases, branch overflow, wraparound, and parity sequences.
H7	A SAR adapter improves focus-quality-per-unit-cost over selected autofocus baselines.	Controlled dataset with disturbance levels, timing, memory, and image metrics.

ID	Hypothesis	Minimum test
H8	An optical or quantum stage reproduces a declared operator with useful error/energy tradeoff.	Physical transfer-function measurement against numerical reference.

15.2 Required ablations

At minimum, experiments should remove or replace one component at a time:

- linear versus log-polar address;
- local zero-set projection versus unconstrained state;
- parity gate versus no gate and stronger integrity code;
- golden-ratio scale versus optimized non-golden scale;
- $\Delta^2\phi$ modulation versus constant branch parameters;
- L-system/BST router versus flat queue or spatial tree;
- self-referential feedback versus open-loop output.

If a simpler ablation performs equally well, the corresponding operator is not yet justified for that application.

15.3 Telemetry contract

Every conforming run should record:

$$\{T_n, u_n, N_n, M_n, \max r_B, \|r_n\|, \text{parity rate}, \text{saturation count}, \text{overflow count}, \text{runtime}\}. \tag{46}$$

This allows failures to be attributed to geometry, control, quantization, resources, or application mismatch rather than hidden by the self-referential loop.

16 Limitations and failure modes

1. **Degenerate field risk.** A literal $F \equiv 0$ eliminates normals and distance information. NHDF requires cell-local nontrivial fields or constraint functions.
2. **Signed-distance ambiguity on non-orientable or self-intersecting surfaces.** Use intrinsic charts, unsigned distance, winding-independent constraints, or local signs; do not assume a global inside/outside label.
3. **Parity blind spots.** One-bit parity misses even fault counts and cannot locate or repair corruption.
4. **Branch explosion.** Recursive rules can exceed memory and time unless node capacity, priority, and recycling are normative.
5. **Gradient singularities.** $\nabla F = 0$ or ill-conditioned fields can make boundary projection unstable; use regularization and fallback logic.
6. **Hybrid chattering.** Frequent parity transitions can destabilize kinematics; use debounce, hysteresis, damping, and rate limits.
7. **Temporal aliasing.** $\Delta^2\phi$ amplifies noise; phase unwrapping, filtering, and sample-rate analysis are required.
8. **Topology is not resilience.** A valid quotient topology does not make hardware immune to radiation, memory exhaustion, numerical error, or corrupted inputs.
9. **Hardware mismatch.** Tensor, ray, or optical units help only when the operator is represented in a form those units actually expose.
10. **Application dependence.** Projection shapes and focus objectives are adapters, not universal outputs

guaranteed by the core.

17 Normative minimum specification

An implementation may call itself an NHDF/PCKF engine only when it satisfies all **MUST** requirements below. **SHOULD** requirements are recommended for reproducibility and safety.

1. It **MUST** define a monotonic logical timeline $T_{\rightarrow}$ and generation semantics.
2. It **MUST** define a log-polar encoding, quantization, saturation, and inverse interpretation.
3. It **MUST** represent "SDF = 0" as non-degenerate local constraints $F_i(z_i, t) = 0$ or explicitly document an equivalent formulation.
4. It **MUST** define the 1-bit parity input, meaning, update, and known blind spots.
5. It **MUST** bifurcate only at declared branch nodes and preserve a declared BST ordering when BST terminology is used.
6. It **MUST** declare how φ_g and $\Delta^2\phi$ affect branching.
7. It **MUST** bound cells, branches, timeline history, and output workspace.
8. It **MUST** declare the analytic sweep and projection operators used to form output.
9. It **MUST** feed a declared output or residual back into the next LUT generation.
10. It **MUST** expose failure when residual, saturation, parity rate, or resource bounds are exceeded.
11. It **SHOULD** provide deterministic replay, reference vectors, and CPU/GPU equivalence tests.
12. It **SHOULD** keep speculative optical, quantum, SAR, AI, or robustness claims separate from core conformance.

An NHDF engine is a bounded causal state machine

$$\mathfrak{N}_{\Delta t} : \mathcal{X} \times \mathcal{L} \to \mathcal{L}$$

whose transition is the closed composition

$$L_{n+1} = \mathcal{U}\Big(L_n, \Pi \circ \mathcal{S}_{\text{cone}} \circ \mathcal{K}_{T_{\rightarrow}} \circ \mathcal{R}_{\text{BST}}^{\varphi_g, \Delta^2\phi_n} \circ \mathcal{P} \circ \mathcal{B}_0 \circ \mathcal{E}_{\text{LP}}(x_n; L_n)\Big),$$

with local validity $F_i(z_i, T_n) = 0$, monotonic logical time, finite resources, explicit parity semantics, and a measurable output residual.

18 Conclusion

The prompt chain can be made coherent by treating it as an operator specification rather than as a list of metaphors. The central shift is distributed zero-set memory: each definable LUT state is valid because it lies on a cell-local implicit boundary. Log-polar jitter chooses the cell; parity produces an event; a golden-ratio and phase-accelerated branch router changes structure; kinematics advance distributed apices; cone sweeps and projections create observables; feedback writes the result into the next generation.

This formulation preserves the source's intended self-reference while making the critical limits explicit. A global field that is identically zero is degenerate. A Klein bottle does not automatically supply a global signed distance in three dimensions. One-bit parity is a gate, not a correction code. Recursive geometry must be bounded. Optical, quantum, SAR, and GPU claims require operator-equivalence tests and application baselines.

The paradigm is therefore ready for the next scientific step: a deterministic reference simulator implementing

Equations (4)–(13), followed by ablations and a bounded GPU port. A positive result would establish NHDF not by the breadth of its vocabulary, but by a reproducible advantage attributable to its ordered, closed composition.

A Symbol table

Symbol	Meaning
x_n	Input sample at logical step n.
$\bar{x}_n, e_n, J_n$	Causal baseline, jitter residual, and scalar jitter magnitude.
ρ_n, θ_n	Log-polar radius and angle.
u_n	Quantized LUT address (k_ρ, k_θ, k_T).
L_n	Complete LUT, cell, branch, timeline, and metadata state.
c_i, q_i	Cell record and packed payload word.
a_i, v_i, α_i	Local apex position, velocity, and acceleration.
φ_g	Golden ratio $(1+\sqrt{5})/2$.
$\phi_i, \Delta^2\phi_i$	Phase and second discrete phase difference.
$T_\rightarrow$	Forward, monotonic logical timeline.
$F_i, \mathcal{M}_i$	Cell-local implicit constraint and its zero-level set.
p_i^{data}	XOR parity of packed payload.
p_i^{topo}	Parity of orientation-reversing topology transitions.
$\widehat{p}_i$	Debounced parity gate used for control.
κ_i	BST ordering key.
$N_{\max}$	Maximum branch-node capacity.
$\mathcal{E}_{\text{LP}}$	Log-polar encoding operator.
$\mathcal{B}_0$	Local boundary embedding/projection operator.
$\mathcal{P}$	Parity reduction operator.
$\mathcal{R}_{\text{BST}}$	BST/L-system branch router.
$\mathcal{K}_{T_\rightarrow}$	Forward kinematic propagation.
$\mathcal{S}_{\text{cone}}$	Analytic cone sweep operator.
Π	Projection/observation operator.
$\mathcal{U}$	Self-referential state update.
$\mathfrak{N}_{\Delta t}$	Complete NHDF transition operator.

B Proof sketches and properties

B.1 Property 1: non-degeneracy

If one global F is identically zero, then every directional derivative is zero wherever F is differentiable. No gradient-based normal or restoring direction can be defined from F. Therefore any implementation using boundary normals must use nontrivial local fields, multiple charts, a constraint Jacobian, or another non-degenerate representation. This establishes the need for Equation (2).

B.2 Property 2: first-order residual contraction

Let $g = \nabla F(z)$ and $z^+ = z - \lambda F(z) g / (\|g\|^2 + \varepsilon)$. Taylor expansion gives

$$F(z^+) = F(z) - \lambda F(z) \frac{\|g\|^2}{\|g\|^2 + \varepsilon} + O(\|z^+ - z\|^2).$$

For small residual and $\varepsilon \ll \|g\|^2$, the leading factor is approximately $1 - \lambda$, giving contraction for $0 < \lambda < 2$.

B.3 Property 3: bounded memory

If all cell and branch allocations are drawn from finite pools, every proposal is accepted only when a free entry exists, and rejected/recycled proposals do not create hidden allocations, then Equation (32) holds by induction. The base state satisfies the bound; each transition preserves it by construction.

B.4 Property 4: parity detection

XOR parity changes under an odd number of bit inversions and remains unchanged under an even number. This follows because XOR is addition modulo 2. Consequently, the parity gate is a valid detector for odd fault count but an incomplete integrity mechanism.

B.5 Property 5: causal ring buffer

Let each physical slot store (g, z), where g is the logical generation. A read for generation g_n rejects any slot tagged $g > g_n$. Even when the physical write pointer wraps, generation tags preserve the partial order and prevent future-state reads. Thus bounded storage is compatible with $T_{\rightarrow}$.

C Traceability to the supplied source

The source is a 30-page exported prompt-and-response transcript. The following mapping shows how its stages are represented in this specification.

Source pages	Source development	Formal destination
1–3	Initial vocabulary and addition of kinematic calculus.	Sections 2–3 and 6: notation, jitter, log-polar addressing, distributed apex, phase, and kinematics.
4–6	Parity-Conditioned Kinematic Foliation, self-referential state, SDF, and feedback.	Sections 4, 5, 7, and 10: canonical chain, local zero sets, parity control, and closure.
7–8	Applications in vegetation, rendering, robotics, and audio.	Application-adapter concept in Section 11; these remain candidate adapters rather than core guarantees.
8–11	Operators and precedence; phase-locked-loop / wavefront-controller framing.	Section 4: typed operator algebra and precedence.
11–15	SAR/auroral mapping and G3 stress behavior.	Sections 7 and 14: parity-rate safe mode and SAR evaluation without immunity claims.
16–18	SDF zero moved from final boundary to the LUT itself; distributed apex; NHDF / Implicit Holographic Co-Processor.	Sections 1, 2, 3, and 5: three-layer identity and distributed zero-set semantics.
19–22	Quantum and optical substrate, $\Delta\Delta\phi$, and 1D timeline directional adherence.	Sections 2, 6, and 13: disambiguated phase acceleration, causal time, and operator-equivalent physical extension.
22–30	Present-day GPU target, kernels, tensor-core priority, compiler and memory profile.	Section 12: semantics-first GPU realization, bounded memory, source-inspired 12 GB example, and explicit hardware caveats.